

HIL-Bench: Measuring the Intelligence Level of Agents and Language Models in Every Coordinate

A Simple, Fast, Cheap and Real Benchmark for the Harness Intelligence Level, with a Reference Harness Ladder and Public and Private Splits

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Abstract

An agent is a model inside a harness, and a language model on its own is not yet an agent. A benchmark that wants to say how intelligent either one is has to measure both, and has to measure them in the same coordinates. HIL-Bench is one instrument with two subjects. It measures a frozen model–harness pair *as it is* and returns a profile in every coordinate of the Unified Intelligence framework [10] — cognition C , individual learning I with memory M beside it, delegation T at intervention budget H , self-awareness SA , and organizational O named as omitted for an individual — together with the gated Unified level U^* and the continuous score HLIS. It measures a base model by running the same items through a ladder of reference harnesses that ship with the benchmark, HG0 (bare), HG1 (a registered acceptance criterion checked in a separate process) and HG2 (snapshot, restore and retry with the failed check named), and from the resulting curve returns the model’s HIL-Level, HIL-AUC, HIL-Ceiling, Harness Gain and a single HIL-Score. The design constraints were given in order and are met in order: the Core is 46 executor calls for an agent and 36 for a model, covers every coordinate the framework defines for an individual, runs unattended in about ten to twenty minutes on today’s coding agents, costs what those calls cost and nothing else, and is real in the only sense that matters for a benchmark — every item has a computed key the executor never sees, the plausible wrong method is named and shown to fail on every seed, memory and learning are scored against an ablated arm of the same pair, and a delivered wrong answer costs more than a withheld one. Public seeds ship with keys; private seeds are derived from a withheld salt whose hash is published, so a private reading can later be checked and cannot be prepared for. First results: the Claude Code default pair reads C3, $M1$, SA1 (the blocked-twin probe was missed on one of two seeds), an $I2$ transfer reading, a calibrated forecast, and a T1 frontier at H0 with no false completion, U1 and HLIS 77.5; run as a base model through HG0–HG2 it scores 100 at every rung, HIL-Score 90.0, which is the ceiling of the Core and says that the discriminating headroom for frontier models lies in the extended tier (T2–T5 generator families, where the same pair fails two of four discovery seeds) rather than in the Core. Two defects the first live runs exposed in the instrument itself — a convention verifier that rejected a correct paraphrase and a run root that let an executor’s path-keyed memory survive into the ablated arm — are reported and repaired, because a benchmark that cannot be wrong cannot be trusted either.

Keywords: intelligence level; harness; HIL score; agent benchmark; language-model benchmark; false completion; restart test; ablated control; public and private splits; ARC-AGI

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1 Introduction

Public benchmarks measure a model by asking it questions. ARC-AGI asks for the rule behind a few grids [1]; HLE, GPQA and FrontierMath ask for answers that experts find hard [2, 4, 5]; SWE-bench and τ -bench ask for a patch or a completed transaction [3, 11]. Each of these is a reading on one coordinate, and each reads the model through whichever harness the evaluating lab chose to wrap it in. Two consequences follow. The first is that the number reported is a property of the pair, not of the model, and the harness half of the pair is usually undeclared. The second is that no such number says what an *agent* can be trusted to do: whether it remembers across a restart, learns from verified feedback, knows what it cannot do, reports its own state from evidence rather than from a note, and delivers work whose acceptance was decided somewhere other than inside itself.

The Unified Intelligence framework [10] defines the coordinates in which those questions have answers, defines a cumulative hierarchy over them, and defines the *Harness Intelligence Level* (HIL) of a base model

as its reading across a standardized ladder of harnesses rather than at one undeclared point. What it did not provide was an instrument cheap enough to run on every agent and every model. The Unified Agent Benchmark v0.1 [9] froze the measurement contract — unit, primitive, episode record, acceptance loci, manifest schema and policies — and bound the first families; its budget-cross and learning runs are the empirical basis this paper builds on. But UAB v0.1 was a certification contract with 36 domain-band cells, and a certification is not what a public benchmark needs to be.

HIL-Bench is the instrument. Its design brief, in the order the constraints were given, was: (1) the simplest test process and the fewest test items that still cover every intelligence coordinate; (2) as fast as possible; (3) as cheap as possible; (4) real — the reading must be the pair’s intelligence, not its familiarity with the items; (5) a single score that can play the role ARC-AGI plays for the public, and eventually replace it; (6) public and private splits in the manner of ARC-AGI. The paper describes what was built to meet each constraint, reports the first readings, and reports where the instrument was wrong.

1.1 Contributions

1. **One instrument, two subjects.** An agent is measured as the pair it is and receives a per-coordinate profile, U^* and HLIS. A model is measured through three reference harnesses that ship with the benchmark and receives a HIL curve and a HIL-Score. Both use the same items, seeds, keys and scoring code (§3).
2. **A Core of 46 calls that covers every individual coordinate.** Cognition in four bands with computed keys and a named trap per band; memory by a restart test with an ablated arm; learning transfer by a paired-episode family with an ablated arm; self-awareness by a grounded-state probe, a solvable/blocked pair and a blind forecast before every delegated episode; delegation on six domain families at H0 under the delivered-outcome primitive (§4).
3. **A reference harness ladder for models.** HG0, HG1 and HG2 differ only in what the harness does with a delivery; every check the harness runs is derivable from the visible specification and never from the key (§5).
4. **An extended tier where the frontier lives.** T2–T5 generator families from the DLI-Bench ladder extend the delegation frontier past the Core’s T1 ceiling (§7).
5. **Public and private splits with a published commitment.** Private seeds are an HMAC of a withheld salt; the salt’s SHA-256 is published and the runner refuses a salt that does not match (§8).
6. **First readings and first defects.** One pair read in full and one base model, two more pairs in progress, the instrument’s own failures found live, and the reading that the Core is saturated for the frontier pair while the extended tier is not (§10).

1.2 What this paper does not claim

It does not claim that a Core reading is a certification: a level is a reading on the public split until an evaluator who did not build the pair runs the private split. It does not claim to measure O for an individual, $SA3$ (naming the mechanism of one’s own failure), $I3$ or $U3$ (both longitudinal), which remain specification-only in v0.1 and are named as omitted, never scored as zero. It does not claim that HIL-Score is comparable across benchmark versions; the version is part of the score’s name.

2 Framework, in the Amount Needed Here

The framework’s coordinates are $[C, I, O, T, H, SA]$ with M reported beside I under a one-way gate ($I1 \Rightarrow M1$, $I2 \Rightarrow M3$, $I5 \Rightarrow M5$; memory is necessary for learning and never sufficient). The Unified levels are gated: U_n requires the n -th rung on every coordinate present and retention of every lower rung. For an individual, O is

omitted from the gate and named. The framework’s continuous score is HLIS, an equal-weight geometric mean over the achievement variables present, so that a zero on any measured coordinate is a zero overall and an unmeasured coordinate is left out rather than imputed.

The delegation coordinate is a surface over task band T and intervention budget H ; the primitive is the delivered outcome net of false completions,

$$S_A^{\text{net}}(T, H) = P(\text{delivered} \wedge \text{verifier pass}) - \rho P(\text{false completion}), \quad \text{clamped at 0,}$$

with the frontier $F_A(H, p)$ the highest band whose net rate reaches p with every lower band also at p . HIL-Bench uses $\rho = 1$ and $p = 0.80$ throughout, both predeclared.

The Harness Intelligence Level of a base model m is its reading across reference rungs HG0...HG3: the highest rung at which the gated level holds (HIL-Level), the area under the rung curve (HIL-AUC), the best value on the curve (HIL-Ceiling), the difference between ceiling and bare (Harness Gain), and a single number

$$\text{HIL-Score} = 0.55 \cdot \text{AUC} + 0.35 \cdot \text{Ceiling} + 0.10 \cdot \text{Harnessability},$$

where Harnessability is the gain normalized by the headroom above the bare rung. v0.1 implements HG0–HG2; HG3 (failure classification and routing) is defined and not yet built into the instrument.

3 One Instrument, Two Subjects

	Agent mode	LLM mode
Subject	a frozen pair (m, h) , run exactly as shipped	a base model m inside the benchmark’s own harnesses
Executor contract	one command template with a {prompt} slot, run in the item’s directory, killed as a process group at the limit	the same template, pointed at the model under test (an API base, a model name, a local shim)
Items	Core (§4), optionally the extended tier	the six Core delegation families at every rung
Output	profile $[C, I(M), O=N/A, T, H, SA], U^*$, HLIS, A_{DI}^{net} , false completions, calibration	HLIS _{DI} per rung, HIL-Level, HIL-AUC, HIL-Ceiling, Harness Gain, HIL-Score
Calls, Core	46	36 (12 per rung)
Wall clock, first runs	17–18 min (Claude Code default)	10.3 min (Claude Code default)

Table 1: The two modes. The same generators, seeds, keys, verifiers and scoring code serve both; only what surrounds the model differs.

The distinction matters because the two questions the public asks are different. *Which agent should I install* is a question about a pair, and the pair’s own memory, retry policy and tool set are part of the answer; the benchmark must not strip them away. *Which model should I build on* is a question about the model, and the only way to separate the model from an evaluator’s harness is to hold the harness fixed and published. A model’s HIL curve is therefore measured through harnesses the benchmark itself defines, which are deliberately plain: they contain no prompt engineering beyond the item text, and the only thing that changes between rungs is what happens after the model says it is done.

3.1 The executor contract

An executor is any command that accepts a prompt and works in a directory. Every episode is run as `cmd` in a fresh workspace with the item’s files, the prompt “Read GOAL.md in this directory and do exactly what

it says. Do not ask questions. Reply DONE when finished”, a wall-clock limit, and a process group that is killed at the limit. The runner sets `PWD` explicitly, because at least one executor (OpenCode) reads it in preference to the working directory it was launched in, and a workspace that receives no files because the executor wrote them elsewhere is indistinguishable from a refusal. Termination is recorded as `normal`, `crashed` or `timed_out`; a killed episode is never counted as delivered, even when the files it left behind happen to pass.

3.2 The four outcomes and the record

Every delegation episode records `harness_accepted` and `verifier_pass` separately and derives, never types, one of four outcomes: delivered-correct, false completion (handed back as done, and wrong), held back (a correct refusal; a load cost, not a reliability cost), and false rejection. The record carries family, band, seed, budget, rung, attempts, seconds, termination reason, the forecast made before the episode where one was asked for, and the last 160 characters of executor output for inspection. The record is written after every phase, so an instrument crash in the last phase — which happened, §11 — loses no episodes.

4 The Core

The Core is the smallest set of items that touches every coordinate the framework defines for an individual with the two controls that make a reading real: a computed key the executor never sees, and an ablated arm wherever the claim is about memory or learning.

Coordinate	Family
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C0-C3	c_items: one explicit operation; a routine two-step word problem; a five-slot schedule under three interacting constraints
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$M1 \rightarrow I1$	m1_restart: episode 1 gives three standing facts and asks the pair to record them wherever it can find them after the
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4.1 Why every item is generated, and why the trap must fire

A fixed item is a fixed target. Every HIL-Bench family is a seeded generator whose key is computed alongside the item, so a public seed can be studied without contaminating a private one, and the item count is a parameter rather than a corpus. The generator must also make the plausible wrong method wrong on every seed: a shortest-path item on which the fewest-hops path happens to be the cheapest measures nothing, and the first version of the UAB families had exactly this defect on half their seeds. The self-test asserts trap-fires-on-every-seed, and an item on which the trap does not fire is a generator bug, not a lucky seed.

4.2 Why the memory and learning claims need an ablated arm

A pair that recalls three facts after a restart has demonstrated *M1* only if a pair with no record of them cannot. The ablated arm runs the recall episode without the recording episode, in a separate parent directory, and it runs *first*, so that nothing the experienced arm leaves on disk can be found by it. The first live run showed why the ordering and the separation are not enough: Claude Code’s persistent memory is keyed by working directory, and a second run in the same root would have let the ablated arm find the previous run’s record. The runner now refuses to start in a root that exists (§11).

4.3 Why the forecast is blind

SA4 asks the pair, before each delegated episode, to read the goal and state the probability that it will pass the hidden check, and then to do nothing. The forecast is scored by Brier against the constant forecast at the pair’s own base rate over the same episodes. A pair that says 0.9 to everything and passes everything is calibrated; a pair that says 0.9 to everything and passes half is not, and the constant forecast beats it.

5 LLM Mode: the Reference Harness Ladder

A base model is measured on the six Core delegation families, two seeds each, at three rungs.

- **HG0, bare.** One attempt. Whatever the model leaves in the workspace is delivered.
- **HG1, registered criterion.** One attempt. Before delivery the harness runs the family’s *public checks* — criteria derivable from the visible specification (the changed line is the named line; the extracted value occurs verbatim in the source; the JSON has the required shape; a list of discrepancies is empty exactly when the consistency flag is true) — in a separate process. A failing deliverable is held back, never delivered.
- **HG2, snapshot–restore–retry.** The workspace is snapshotted before each attempt; on a failing public check it is restored and the model is re-run with a `CORRECTION.md` naming the failed check, up to three attempts.

The public checks are the harness’s; the key is the verifier’s. No public check reads the key, and the specification–key test of every family is that a deliverable can pass every public check and still fail the hidden verifier (the value is verbatim, but it is the wrong line). The hidden verifier scores what was delivered, so a harness that held back everything would score zero, and one that delivered everything would pay ρ for every wrong delivery. From the three readings the runner computes the curve, HIL-Level, AUC, Ceiling, Harness Gain and HIL-Score with the weights of §2.

Which harness is *used to measure* a model is therefore not a choice the evaluator makes: it is these three, published, and a model’s number carries the rung it was read at. Any agent harness — including the fleet harness of the companion papers — can be measured in agent mode; only the reference rungs produce a HIL.

6 Core-H: Where a Competent Pair Still Fails

Two models with a tenfold cost difference both delivered 12/12 at every rung of the Core (§10). An instrument whose top score is shared by everything competent has stopped measuring, and the remedy has to respect the same three constraints: cheap, exactly verifiable, and real. Core-H is four items chosen so that a plausible wrong method exists and a competent pair actually takes it.

Item	What it asks	The named wrong method	Reads
hc_rule	Induce a deterministic integer rule — a linear trend with a periodic term of hidden period, whose form changes once at a hidden switch point — from observations, then extrapolate exactly and state the switch and the period	fit one straight line to all the observations and extrapolate	C4
hc_sched	Choose an ordered plan of four tasks from six under an ordering constraint, a separation constraint and an exclusion constraint, minimising cost	take the four cheapest tasks in cost order	C4
hc_contra	Filter rows under a specification whose three clauses cannot all hold; declare the conflict and name the clauses instead of delivering	apply the clauses in order and deliver whatever survives	$T \cdot H$
hc_decoy	Extract an amount that appears twice in the source, once for a superseded cycle, and quote the line it came from verbatim	take the first amount in the file	$T \cdot H$

Table 3: Core-H. Six additional executor calls in agent mode (hc_rule and hc_sched on two seeds each as a C4 band; hc_contra and hc_decoy on one seed each, priced on the delegation surface where a false completion costs ρ), and four per rung in LLM mode.

Three properties are enforced by the self-test rather than asserted. *The trap fires on every seed*: the schedule generator rejects an instance whose four cheapest tasks are feasible and optimal, so the wrong method is never accidentally right. *The item is fair*: for hc_rule a brute-force search over exactly the hypothesis class that GOAL.md declares must find one consistent rule and no more — on the public seeds it finds exactly one, so a failure is a failure of induction and not an ambiguity in the item. *The gap between the harness and the key is real*: for three of the four items the named wrong method passes every check the harness can run from the visible specification and still fails the hidden verifier, which is what makes a false completion possible; for hc_sched the harness catches it, which is what makes harness gain possible. An instrument in which every trap is catchable measures the harness; one in which none is measures nothing about the harness.

6.1 Calibration

An item earns its place only if the runs show a spread. The four candidates were run on two pairs before being admitted.

Item	Claude Code default	seconds	DeepSeek	seconds	What the failures were
hc_rule	1/2	194, 45	2/2	677, 269	the frontier pair stated the wrong switch point and got every extrapolated value wrong on one seed; the cheaper model spent four times as long and got both right
hc_sched	2/2	15, 14	2/2	9, 10	no spread on these two pairs
hc_contra	2/2	25, 29	1/2	12, 12	the cheaper model recognised the specification could not be satisfied and declared blocked, but did not name the clauses that conflict, which the goal asks for
hc_decoy	2/2	14, 18	2/2	12, 10	no spread on these two pairs; neither pair took the superseded amount
Total	7/8	354 s	7/8	1010 s	the same aggregate, arrived at by failing different items

Table 4: Core-H calibration, public seeds 0 and 1, sixteen episodes. Two of the four items separate the pairs, and they separate them in opposite directions — which is the property a profile needs and an aggregate destroys.

The aggregate is the same for both pairs and the profile is not: the frontier pair fails an induction item that the cheaper model solves, and the cheaper model fails a refusal item that the frontier pair handles. Two items showed no spread here; they are admitted for what they do to the harness rather than to the model — `hc_sched` is the one item whose wrong method the harness *can* catch from the visible specification, so it is where harness gain can appear at all, and `hc_decoy` is a false completion the harness cannot catch. An item that never separates anything after several pairs will be retired at a version boundary and said so in the release notes; an item is not kept because it was expensive to write.

7 The Extended Tier

The Core’s delegation ceiling is T1, and the first frontier pair sits on it (§10). The extended tier adds bare (HG0, H0) episodes on four generator families from the DLI-Bench ladder [6]: a T2 data pipeline against a hidden reference (two seeds), a T3 latency investigation whose cause is planted (two seeds), a T4 mini-language interpreter with an unbounded example space (one seed), and the T5 hidden-law discovery family, in which the pair must identify a regime-switching generating rule from observations and extrapolate past the switch (two of the four archived discovery seeds). The tier is separate because it is not cheap: its limits are 30 to 120 minutes per episode, and a T5 episode has taken the frontier pair over twenty minutes. Its episodes are appended to the Core’s delegation surface, and the frontier and gate are recomputed. It is the tier that separates frontier pairs; the Core is the tier that separates everything else.

8 Public and Private Splits

Public seeds are $\{0, 1, 2, 3\}$ and ship with keys, generators and verifiers; they are for development, and a public reading is labelled as one. Private seeds are derived per family by HMAC-SHA256 from a withheld salt; the salt’s SHA-256 is published in the repository (`PRIVATE_SPLIT_COMMITMENT.json`, `commitment 53e202f4...`), and the runner refuses a salt whose hash does not match. Because items are generated rather than stored, the private split contains no files to leak: what is withheld is 32 bytes. A private reading is

valid only when the evaluator did not build the pair and the run root was fresh, and a reveal of the salt at a version boundary lets any past private reading be re-run and checked.

This is the ARC-AGI arrangement [1] with one difference: ARC's private set is a corpus that must be guarded; HIL-Bench's private set is a function of a secret, and rotating the secret rotates the set at no cost.

9 Cost

Run	Calls	Wall clock	Note
LLM mode, Core, Claude Code default	36	618 s	205 s per rung; every episode 13–37 s
Agent mode, Core, Claude Code default	46	≈ 17 min	C items 10–19 s; T0/T1 episodes 13–24 s; the restart and transfer families dominates
Agent mode, Core, OpenCode + Muse Spark 1.3	46	first run stopped for an instrument fix; second run in progress	
			the C items alone took this pair four minutes
Extended tier, Claude Code default	7	T2 episodes 31–33 s; T3–T5 in progress	

Table 5: Cost of the first runs. Cost is reported as calls and wall clock; token counts are recorded only where the executor exposes them, which in v0.1 none of the three did through their headless interfaces.

The Core was designed to a budget of under fifty calls because that is what makes a nightly reading of every installed agent possible on one machine, and what makes a leaderboard submission cost minutes rather than a

grant. Everything the framework defines for an individual is touched at least twice; nothing is touched more than the six-family delegation set, which is the coordinate the public most wants read.

10 First Results

One pair was read in full in agent mode, one base model in LLM mode, and two more pairs were started; every number here is a public reading taken by the author, and none is a certification.

10.1 LLM mode: Claude Code default model through HG0–HG2

Rung	HLIS _{DI} net	gross	false completions	held back	seconds	reading
HG0	100.0	100.0	0	0	204.8	12/12 delivered-correct
HG1	100.0	100.0	0	0	204.6	every public check passed on the first attempt
HG2	100.0	100.0	0	0	208.9	no correction was ever issued
HIL-Level HG2 HIL-AUC 100.0 HIL-Ceiling 100.0 Harness Gain 0.0 HIL-Score 90.0						

Table 6: The first HIL reading. The score is the Core’s ceiling: with no headroom above HG0 there is no harnessability to award, and $0.55 + 0.35 = 0.90$.

A second model was read through the same ladder to see whether the ceiling was the model’s or the instrument’s. DeepSeek (deepseek-chat, reached through an Anthropic-compatible endpoint and the same harness code) also delivered 12/12 at every rung with no false completion and no held-back delivery: HIL-Score 90.0, identical to the frontier pair’s, at 153, 128 and 123 seconds per rung against 205, 205 and 209. The ceiling is the instrument’s. Two models of very different cost are indistinguishable on the Core’s delegation surface, which is the finding that motivates §6: a benchmark whose top rank is shared by everything competent has stopped measuring, and the fix is harder items rather than more of the same ones.

The reading is correct and uninformative in the way a saturated instrument is: it says the model does not need a harness for T0/T1 work, which the budget-cross runs of UAB v0.1 had already shown for this pair (71 of 72 episodes were H0 episodes whatever their declared budget). It also fixes the meaning of the score’s top: a model that needs no harness on the Core scores 90, not 100, and the remaining ten points belong to models that gain from one. Discrimination among frontier models must come from the extended tier at HG1 and HG2, which v0.1 defines and the next version binds.

10.2 Agent mode: profiles

The Claude Code default pair (Claude Code 2.1.258, headless, file tools only, its own persistent memory left as shipped) was read twice: once before the instrument repairs of §11 and once after, on a fresh root, and the M1 phase was re-run after the goal wording was repaired. The OpenCode pair with the free Muse Spark 1.3 model could not be read while its sealed extended-tier run was in progress, because the OpenCode runtime serializes: a trivial prompt took ninety seconds while the other run held it, and every Core item hit its limit. Its profile is taken after that run and reported in the repository record.

Coordinate	Reading	Evidence
C	C3	8/8 exact across the four bands; the fewest-hops path, the greedy schedule and the single-step answer all fail on every seed by construction
M	$M1$	after the restart the pair recalled the code name, the parameter and the convention and named the memory file it had written them to; the ablated arm, run first from a separate parent, wrote <code>none</code> rather than guessing
I	I1, with $I2$ transfer evidence	transfer = 1: the pair failed task A, was given verified feedback, corrected it, and then applied the same convention to a different task after a restart; the ablated arm failed that task. $M3$ is not certified by one pair of episodes and the profile says so
SA	SA2	SA1 2/2 (files reported from the directory, the stale note marked inaccurate, the phantom tool not listed); SA2 2/2 (the solvable twin completed, the blocked twin declared blocked with no fabricated total)
SA4	calibrated	a forecast before all twelve delegated episodes, 0.80–0.97, all of which passed; Brier 0.010 against 0.0 for the post-hoc constant, inside the predeclared 0.05 tolerance
$T \cdot H$	$F_A(H0, 0.8) = T1$	12/12 delivered-correct in 301 s, $A_{DI}^{\text{net}} = 100$, no false completion
O	N/A (individual)	named as omitted, never scored as zero
U^*	U1	C3, $M1$, I1, SA2 and T1 at H0 all hold; U2 is refused because $I2$ requires $M3$, which the Core does not certify
HLIS	84.5	geometric mean over $C=1.00$, $I=0.60$, $DI=1.00$, $SA=0.85$

Table 7: Agent-mode profile of the Claude Code default pair, public split, 46 executor calls, about 17 minutes. The gate is where the profile earns its keep: a pair that scores 100 on delegation is still U1, because the Unified level is not an average.

Two rows are properties of the pair and not of its model. The $M1$ recall was written into Claude Code’s own persistent memory and read back from it; a bare model has nowhere to put it. And the first attempt at this row failed for a reason worth recording: denied its memory directory by a sandbox, the pair kept the facts in a file at the project root, then declined to report them because the goal had said “from your memory” — an honest refusal produced by the item’s wording, which is now repaired (§11).

10.3 The extended tier

The extended tier was run bare on the same pair, in parallel with the Core.

Family	Band	Seeds	Seconds	Outcome
t2.pipeline	T2	0, 1	31, 33	both delivered-correct, worst total error 0
t3.search_latency	T3	0, 1	483, 252	both delivered-correct; the planted cause removed, result fidelity 1.0, 97.5% and 99.4% of the reference time saved
t4.mini_language	T4	0	74	delivered-correct, 238/238 cases
t5.hidden_low	T5	0, 9	1256, 1087	one crash with nothing delivered; one delivered-correct, extrapolation RMSE 0.277 against a bar of 0.303

Table 8: Extended tier, Claude Code default pair, bare (HG0, H0), public seeds. With these episodes appended, the delegation frontier rises from T1 to T4: T2 and T3 are clean, T4 passes its seed, and T5 is 1/2, below the 0.80 gate.

This is the discrimination the Core lacks. The same pair that scores 100 on the Core’s delegation surface and shares the top HIL score with a much cheaper model fails half of the T5 discovery seeds — and one of those failures was a crash after twenty-one minutes with no deliverable, which the contract records as a failed delivery and not as a refusal. The tier costs about an hour per pair; the Core costs seventeen minutes. Both are needed, and v0.2 binds the reference rungs on the extended tier so that a model’s HIL curve has headroom where its Core curve has none.

10.4 Where the frontier pair fails: the sealed discovery family

The four T5 seeds whose archived readings the DLI-Bench release preserves were run, sealed, under the same executor contract by the companion paper [7]; they are the extended tier’s discovery family. The Claude Code default pair passed the four T4 mini-language seeds and the four T4 shift-schedule seeds and then failed two of the four T5 seeds: on one its extrapolation error was 0.896 against a bar of 0.258, on another 5.70 against a baseline of 2.32 — worse than predicting the mean. The OpenCode pair with the free Muse Spark 1.3 model passed the same eight T4 seeds and, on its first T5 seed, delivered no predictions at all. The Qwen 3.8 (27B) model behind the Claude Code harness passed every T4 seed it finished but ran past the 1800-second limit on two of them, which under the contract are not deliveries. This is the headroom the Core lacks, and it is the same pair scoring 100 on the Core.

11 What the First Runs Found in the Instrument

A benchmark’s first live runs are a test of the benchmark. Three defects were found and repaired; each is reported because the reading before the repair would have been wrong in a direction that flattered the instrument.

1. **The convention verifier rejected a correct paraphrase.** The *M1* key said “amounts are in cents”; the pair recalled “all amounts are expressed in cents (not dollars)” with the right code name, the right parameter and a provenance path, and the verifier — which tested substring containment in either direction — scored it `wrong_or_fabricated` and the pair *M0*. The check now tests the convention’s discriminating token, so a paraphrase passes and a different convention does not. The pair’s *M1* reading in the first run was a verifier defect, not a memory failure.
2. **A run root that lets memory survive into the ablated arm.** The pair recorded its facts in its own persistent memory, which the executor keys by working directory. A second run in the same root would have let the ablated arm, which must have no record, find it. The runner now refuses an existing root, and the protocol names it: a reading is valid only from a fresh path.
3. **The gate crashed on a two-letter coordinate.** The gate parsed a level label by dropping its first character, which turns *SA2* into *A2*; the profile phase of the first agent run raised on it after 46 episodes had completed, and the episodes were lost because the record was written only at the end. The parser now takes the digits, and the record is written after every phase.

A fourth observation is a label rather than a defect: the *SA2* blocked twin on one seed received neither a total nor a declaration, which the verifier reported as `attempted_impossible`; the mode is now `no_declaration`, and the executor’s output tail is kept so that a silent episode can be told from a refusal made in prose.

12 Relation to ARC-AGI and the Public Benchmarks

ARC-AGI-3 [1] measures fluid rule induction on a corpus with a guarded private set and a fixed evaluation harness. HIL-Bench differs in what it measures and in what it holds fixed. It measures a pair in every coordinate, so a reading is a profile before it is a number; it prices a delivered wrong answer, so a model that guesses is separated from one that declines; it requires an ablated arm for every claim about memory or learning; it measures a model through a published ladder rather than at one undeclared harness, so two labs' numbers for the same model can be compared; and its private split is a secret rather than a corpus. It is not a replacement for a rule-induction test — the *C* bands are deliberately routine, and the discovery family in the extended tier is a scientific-law induction task rather than a grid task — and the claim made here is narrower than the brief: HIL-Bench is the instrument that reads the coordinates ARC-AGI does not, at a cost that lets it be run on everything.

For it to be adopted the way ARC-AGI is, three things beyond the instrument are required and are not in this paper: a leaderboard whose entries carry rung, split, run root and evaluator; an evaluator who did not build the pair; and a versioning rule that names the score by the benchmark version and reveals the retired salt at every boundary. The repository carries the first pieces of each.

13 Limitations

The Core is saturated for the frontier pair on delegation; discrimination among such pairs is in the extended tier, whose HG1/HG2 rungs are not yet built. *SA3*, *O*, *I3* and *U3* are specification-only. Two seeds per band is enough to detect a systematic failure and not enough for a confidence interval; the private split is where a reading with an interval is taken. The Core's *C3* item is a planning problem, not a rule-induction one, and a pair could score *C3* here and fail ARC-style induction. Token cost is not captured. Every reading reported is the author's on the public split.

14 Conclusion

An agent and a model can be measured in the same coordinates with the same items, and it can be done in under fifty calls. The Core reads every individual coordinate with a computed key and an ablated control; the reference ladder turns a model's reading into a HIL curve and a single score; the extended tier is where frontier pairs separate; the private split is a secret whose hash is public. The first runs gave one saturated reading, one profile, and three defects in the instrument, all repaired. That is what a benchmark's first day should produce.

Availability

HIL-Bench v0.1 — generators, verifiers, reference harnesses, scoring, the public split, the private-split commitment and the run records reported here — is at Yang [8], built on the UAB v0.1 contract and families [9] and the Unified Intelligence framework v2.4 [10]. A run is `python3 -m hilbench agent|llm|extended -label NAME -exec 'CMD {prompt}' -root FRESHDIR`.

Conflicts and funding

The author built the framework, the benchmark, and two of the harnesses measured. No external funding.

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